data visualisation

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LOAD THE NECESSARY PACKAGE

```r
library("dplyr")
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library("lubridate")
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```r
library("ggplot2")
library("tidyr")
```

SET THE WORKING DIRECTORY

```r
setwd("C:/Users/Admin/Desktop/DML")
```

LOAD THE WORKING DIRECTORY

```r
library(readr)
df<- read_csv("StudentsPerformance.csv")
```

```
## Rows: 1000 Columns: 8
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (5): gender, race/ethnicity, parental level of education, lunch, test pr...
## dbl (3): math score, reading score, writing score
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
View(df)
```

LOAD THE NECESSARY PACKAGE

```r
# First, load the dataset
students <- read.csv("StudentsPerformance.csv")

# Fix column names (remove spaces)
names(students) <- make.names(names(students))

# Look at the data
head(students)
```

```
##   gender race.ethnicity parental.level.of.education        lunch
## 1 female        group B           bachelor's degree     standard
## 2 female        group C                some college     standard
## 3 female        group B             master's degree     standard
## 4   male        group A          associate's degree free/reduced
## 5   male        group C                some college     standard
## 6 female        group B          associate's degree     standard
##   test.preparation.course math.score reading.score writing.score
## 1                    none         72            72            74
## 2               completed         69            90            88
## 3                    none         90            95            93
## 4                    none         47            57            44
## 5                    none         76            78            75
## 6                    none         71            83            78
```

```r
colnames(data)
```

```
## NULL
```

##Question 1: Line Graph - Average Scores by Parental Education

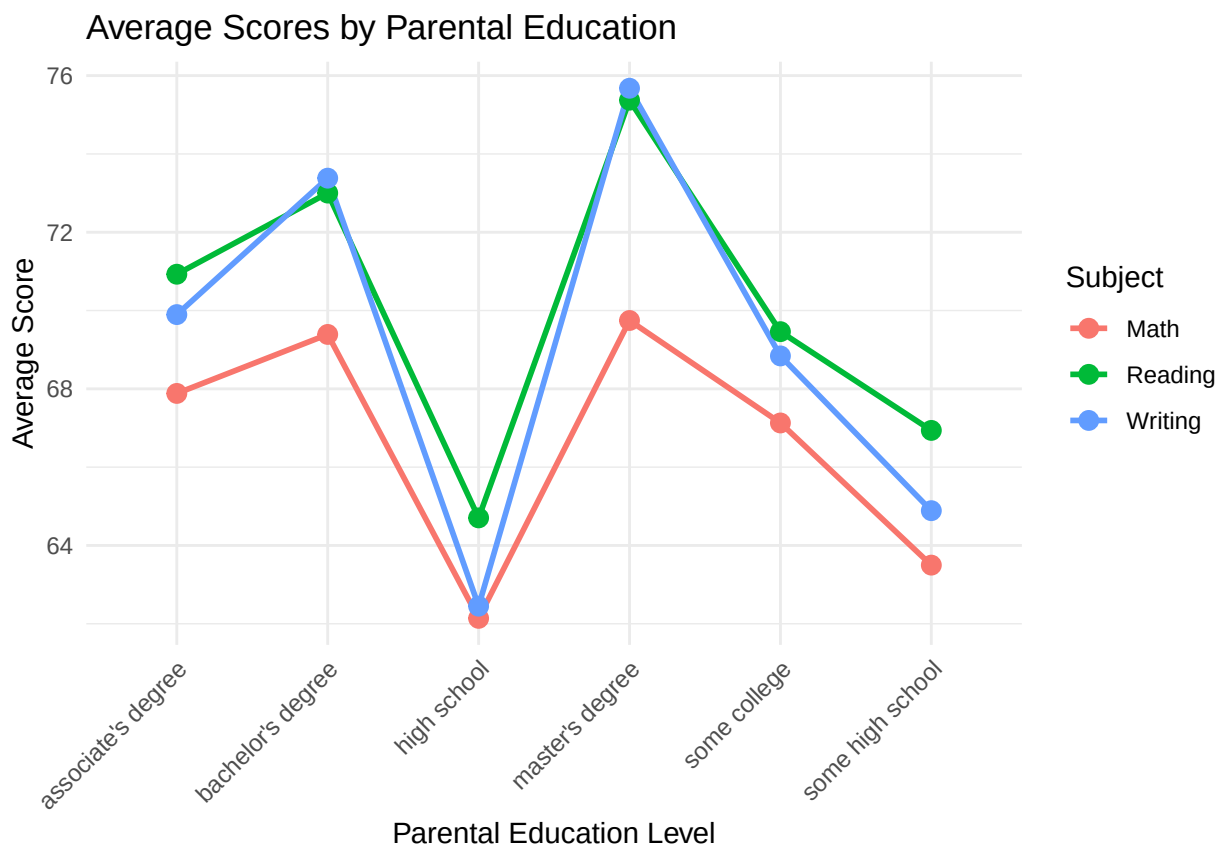
```r
# Q1: Calculate averages
q1_data <- summarise(group_by(students, parental.level.of.education),
                     Math = mean(math.score),
```

```r
                      Reading = mean(reading.score),
                      Writing = mean(writing.score))

# Convert to long format
q1_long <- pivot_longer(q1_data,
                        cols = c("Math", "Reading", "Writing"),
                        names_to = "Subject",
                        values_to = "Average_Score")

# Create plot
ggplot(q1_long, aes(x = parental.level.of.education,
                    y = Average_Score,
                    group = Subject,
                    color = Subject)) +
  geom_line(size = 1) +
  geom_point(size = 3) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Average Scores by Parental Education",
       x = "Parental Education Level",
       y = "Average Score")
```

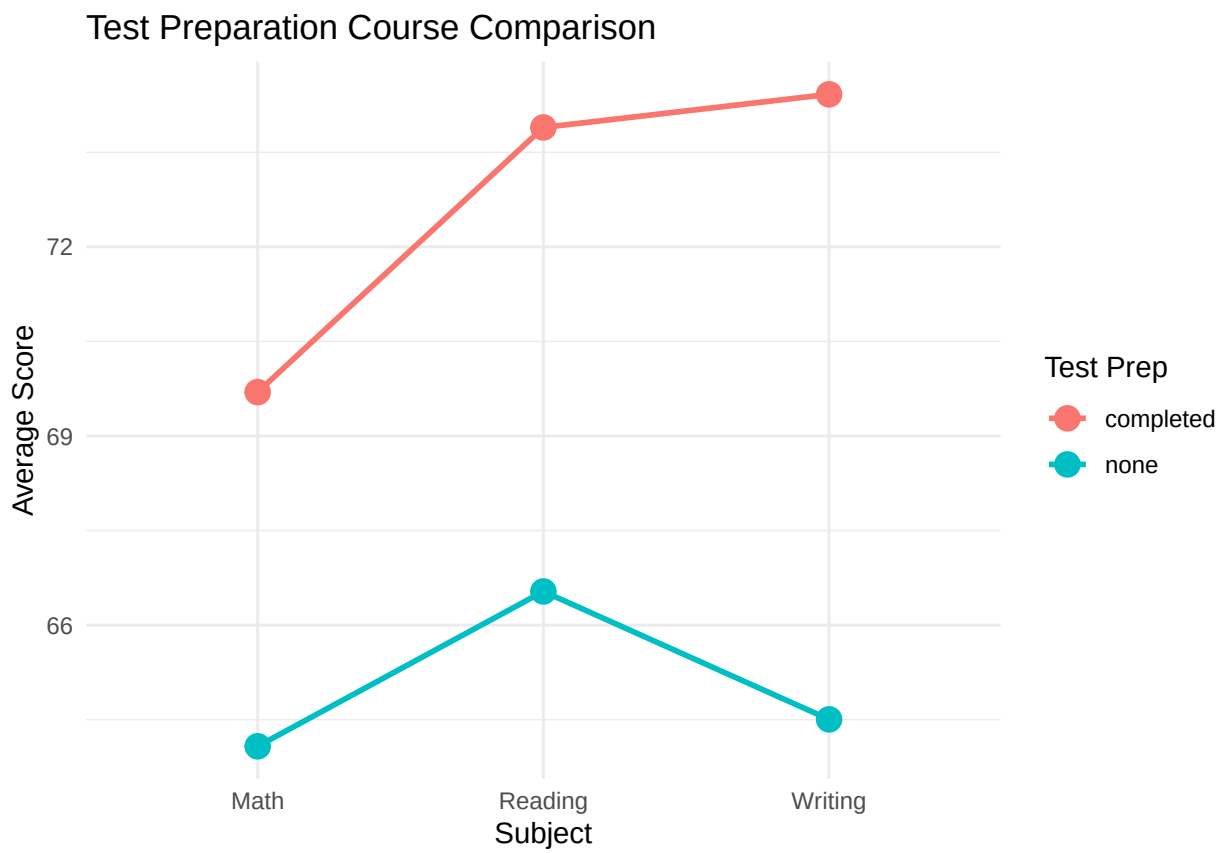
```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



Q2: Calculate averages by test prep

```
# Calculate averages by test prep
q2_data <- students %>%
  group_by(test.preparation.course) %>%
  summarise(
    Math = mean(math.score),
    Reading = mean(reading.score),
    Writing = mean(writing.score)
  ) %>%
  pivot_longer(cols = c(Math, Reading, Writing),
               names_to = "Subject",
               values_to = "Average_Score")

# Create line graph
ggplot(q2_data, aes(x = Subject,
                    y = Average_Score,
                    group = test.preparation.course,
                    color = test.preparation.course)) +
  geom_line(size = 1) +
  geom_point(size = 4) +
  theme_minimal() +
  labs(title = "Test Preparation Course Comparison",
       x = "Subject",
       y = "Average Score",
       color = "Test Prep")
```




```r
# Check if preparation helps consistently
q2_comparison <- q2_data %>%
  pivot_wider(names_from = test.preparation.course,
              values_from = Average_Score) %>%
  mutate(Improvement = completed - none)

print("Score improvement from test prep:")
```

```
## [1] "Score improvement from test prep:"
```

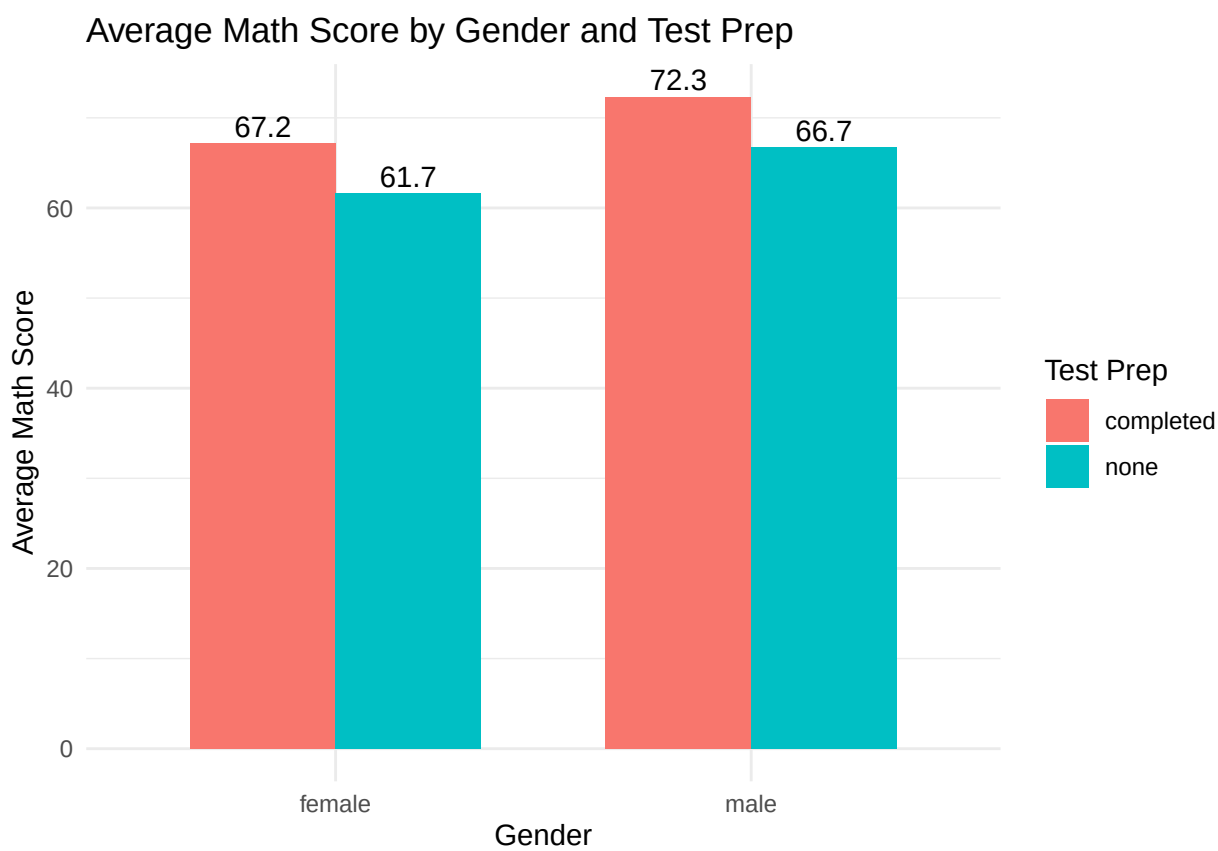
```r
print(q2_comparison)
```

```
## # A tibble: 3 x 4
##   Subject completed  none Improvement
##   <chr>       <dbl> <dbl>       <dbl>
## 1 Math         69.7  64.1        5.62
## 2 Reading      73.9  66.5        7.36
## 3 Writing      74.4  64.5        9.91
```

Question 3: Grouped Bar Chart - Math Score by Gender and Test Prep

```r
# Calculate average math scores
q3_data <- students %>%
  group_by(gender, test.preparation.course) %>%
  summarise(avg_math = mean(math.score), .groups = 'drop')

# Create grouped bar chart
ggplot(q3_data, aes(x = gender,
                    y = avg_math,
                    fill = test.preparation.course)) +
  geom_bar(stat = "identity", position = "dodge", width = 0.7) +
  geom_text(aes(label = round(avg_math, 1)),
            position = position_dodge(0.7),
            vjust = -0.3) +
  theme_minimal() +
  labs(title = "Average Math Score by Gender and Test Prep",
       x = "Gender",
       y = "Average Math Score",
       fill = "Test Prep")
```



```r
# Find highest group
q3_highest <- q3_data %>%
  arrange(desc(avg_math)) %>%
  slice(1)

print("Group with highest math score:")
```

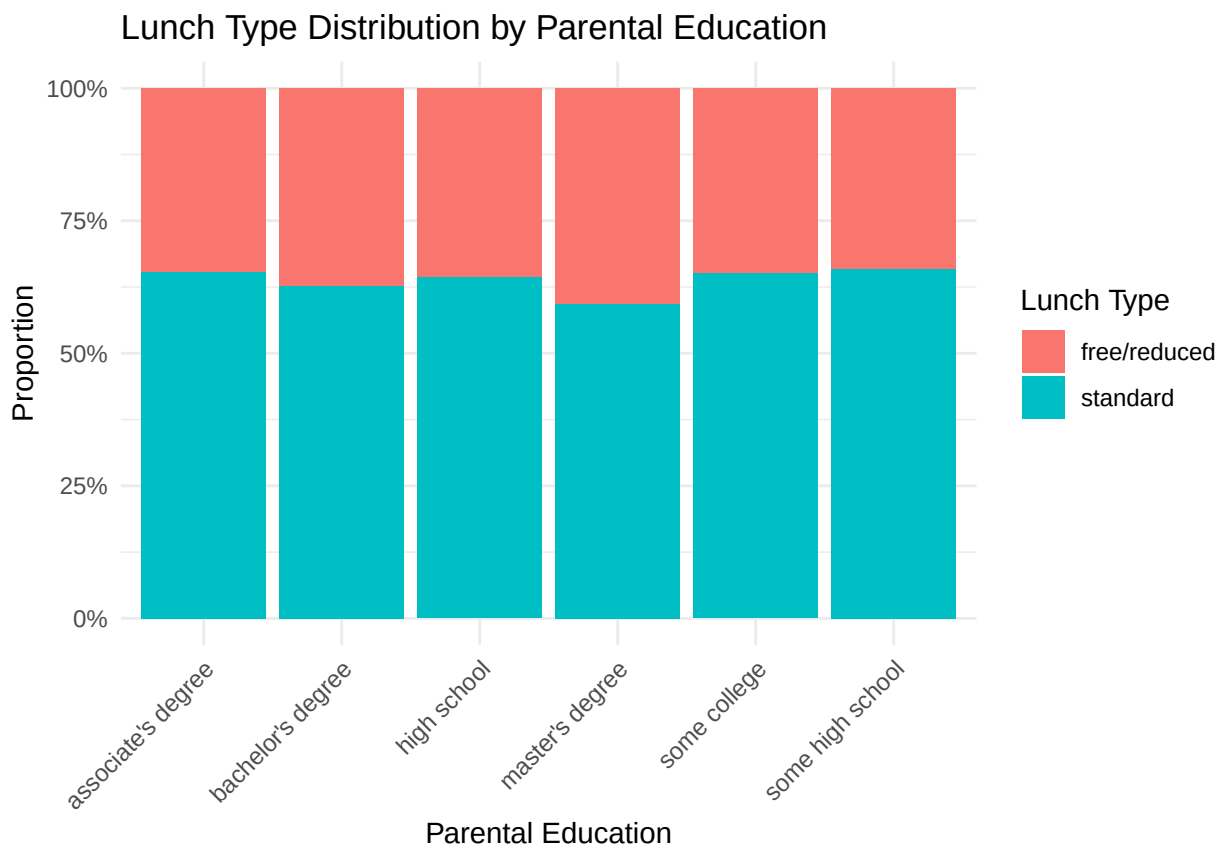
```
## [1] "Group with highest math score:"
```

```r
print(q3_highest)
```

```
## # A tibble: 1 x 3
##   gender test.preparation.course avg_math
##   <chr>  <chr>                      <dbl>
## 1 male   completed                   72.3
```

```r
# Count students by parental education and lunch type
q4_data <- students %>%
  group_by(parental.level.of.education, lunch) %>%
  summarise(count = n(), .groups = 'drop') %>%
  group_by(parental.level.of.education) %>%
  mutate(proportion = count / sum(count))

# Create stacked bar chart
ggplot(q4_data, aes(x = parental.level.of.education,
                    y = proportion,
                    fill = lunch)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Lunch Type Distribution by Parental Education",
       x = "Parental Education",
       y = "Proportion",
       fill = "Lunch Type") +
  scale_y_continuous(labels = scales::percent)
```




```r
# Find highest free/reduced lunch group
q4_highest <- q4_data %>%
  filter(lunch == "free/reduced") %>%
  arrange(desc(proportion))

print("Highest free/reduced lunch proportion:")
```

```
## [1] "Highest free/reduced lunch proportion:"
```

```r
print(head(q4_highest, 1))
```

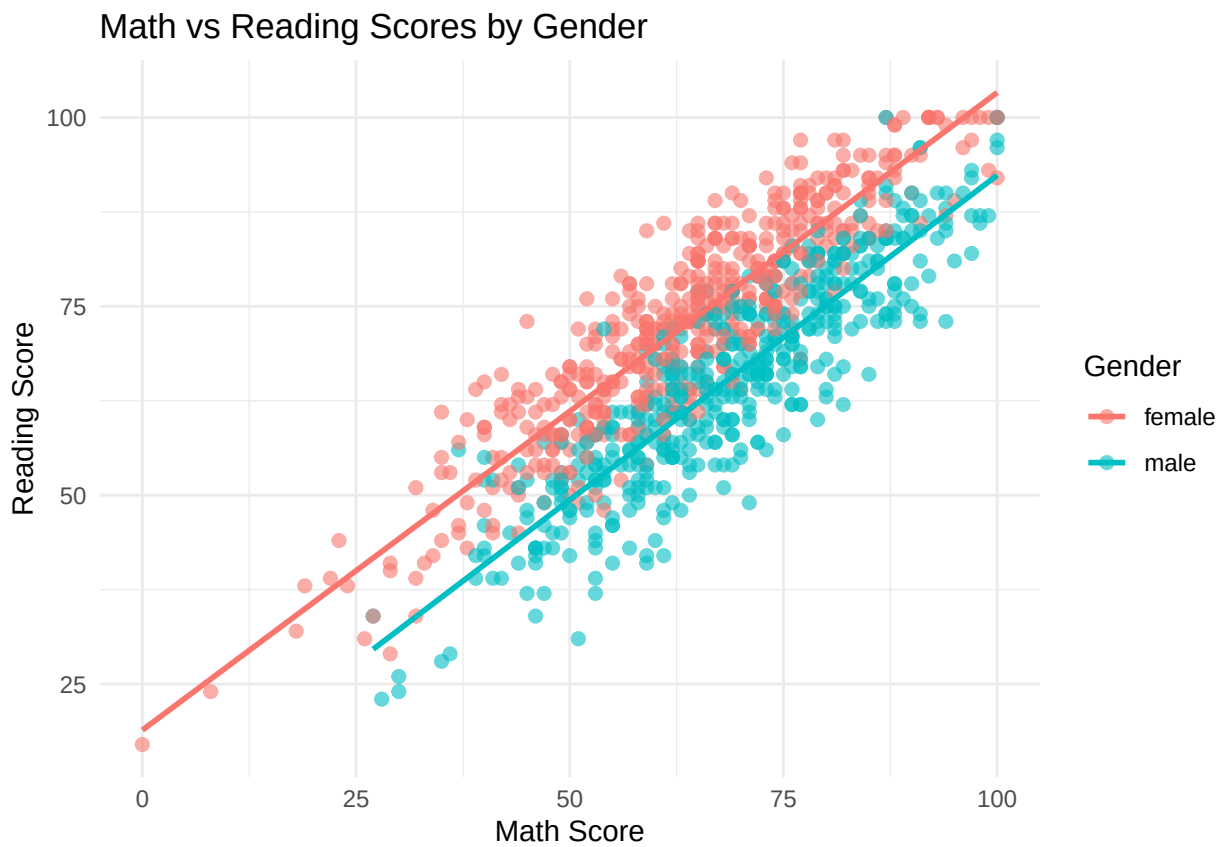
```
## # A tibble: 1 x 4
## # Groups:   parental.level.of.education [1]
##   parental.level.of.education lunch        count proportion
##   <chr>                       <chr>        <int>      <dbl>
## 1 master's degree             free/reduced    24      0.407
```

##Question 5: Scatter Plot - Math vs Reading Score

```r
# Create scatter plot
ggplot(students, aes(x = math.score, y = reading.score,
                     color = gender)) +
  geom_point(alpha = 0.6, size = 2) +
  geom_smooth(method = "lm", se = FALSE) +
```

```r
  theme_minimal() +
  labs(title = "Math vs Reading Scores by Gender",
       x = "Math Score",
       y = "Reading Score",
       color = "Gender")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Calculate correlations by gender
q5_correlations <- students %>%
  group_by(gender) %>%
  summarise(
    correlation = cor(math.score, reading.score)
  )

print("Correlation by gender:")
```

```
## [1] "Correlation by gender:"
```

```r
print(q5_correlations)
```

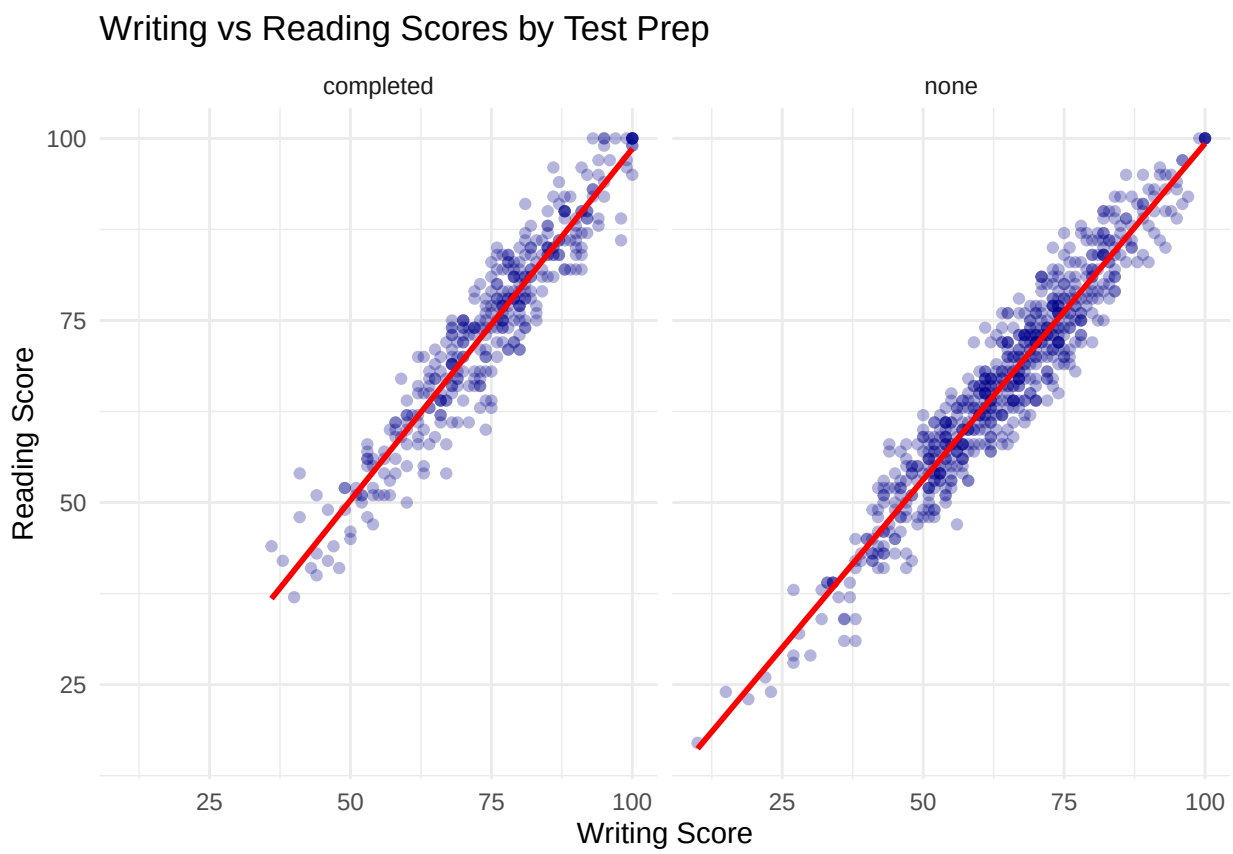
```
## # A tibble: 2 x 2
##   gender correlation
```

```
##   <chr>        <dbl>
## 1 female       0.909
## 2 male         0.885
```

##Question 6: Scatter Plot - Writing vs Reading Score (Faceted)

```r
# Create faceted scatter plot
ggplot(students, aes(x = writing.score, y = reading.score)) +
  geom_point(alpha = 0.3, color = "darkblue") +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  facet_wrap(~test.preparation.course) +
  theme_minimal() +
  labs(title = "Writing vs Reading Scores by Test Prep",
       x = "Writing Score",
       y = "Reading Score")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Calculate correlations by test prep
q6_correlations <- students %>%
  group_by(test.preparation.course) %>%
  summarise(
    correlation = cor(writing.score, reading.score)
  )

print("Correlation by test prep:")
```

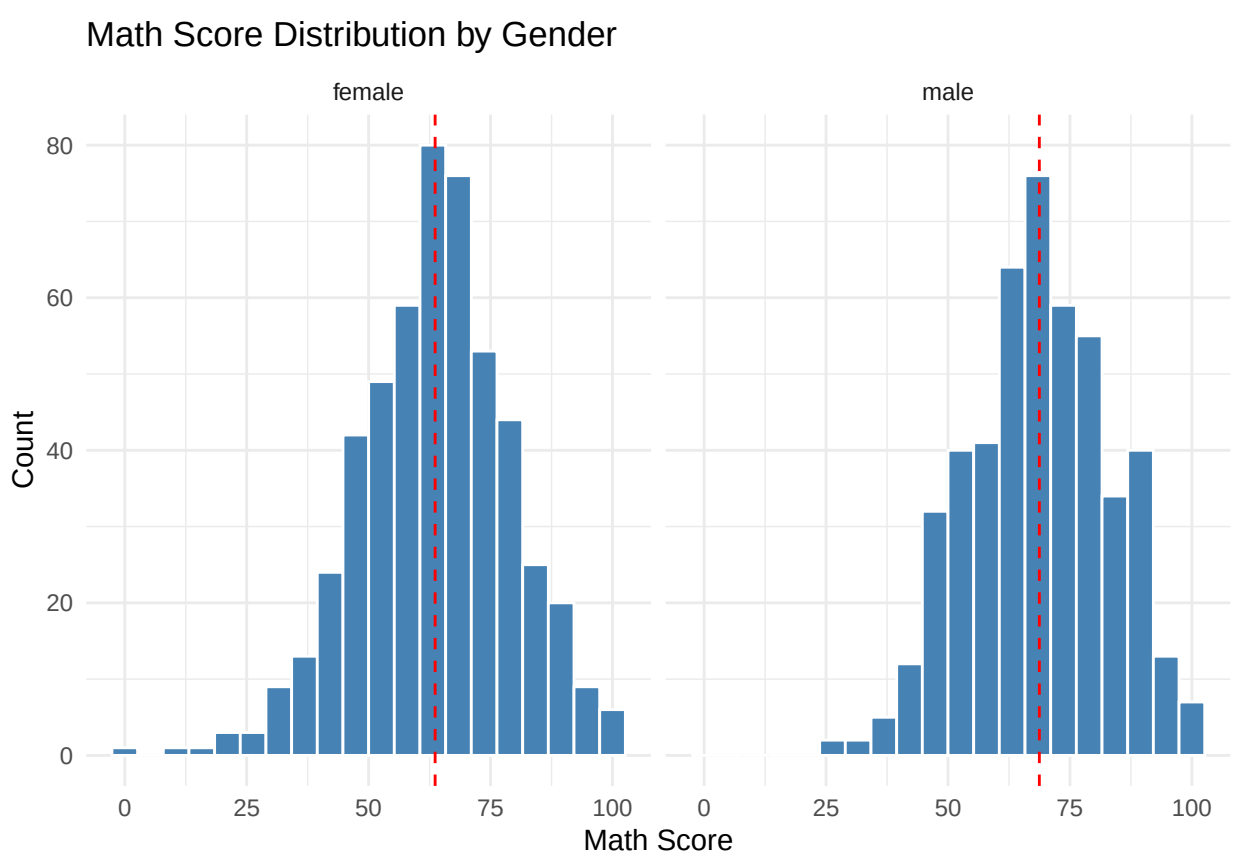
```
## [1] "Correlation by test prep:"
```

```r
print(q6_correlations)
```

```
## # A tibble: 2 x 2
##   test.preparation.course correlation
##   <chr>                        <dbl>
## 1 completed                    0.947
## 2 none                         0.957
```

##Question 7: Histogram - Math Score Distribution by Gender

```r
# Create faceted histogram
ggplot(students, aes(x = math.score)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  facet_wrap(~gender) +
  theme_minimal() +
  labs(title = "Math Score Distribution by Gender",
       x = "Math Score",
       y = "Count") +
  geom_vline(data = students %>%
               group_by(gender) %>%
               summarise(mean_math = mean(math.score)),
             aes(xintercept = mean_math),
             color = "red", linetype = "dashed")
```



```r
# Summary statistics
q7_stats <- students %>%
  group_by(gender) %>%
  summarise(
    Mean = mean(math.score),
    Median = median(math.score),
    SD = sd(math.score),
    Min = min(math.score),
    Max = max(math.score)
  )

print("Math score statistics by gender:")
```

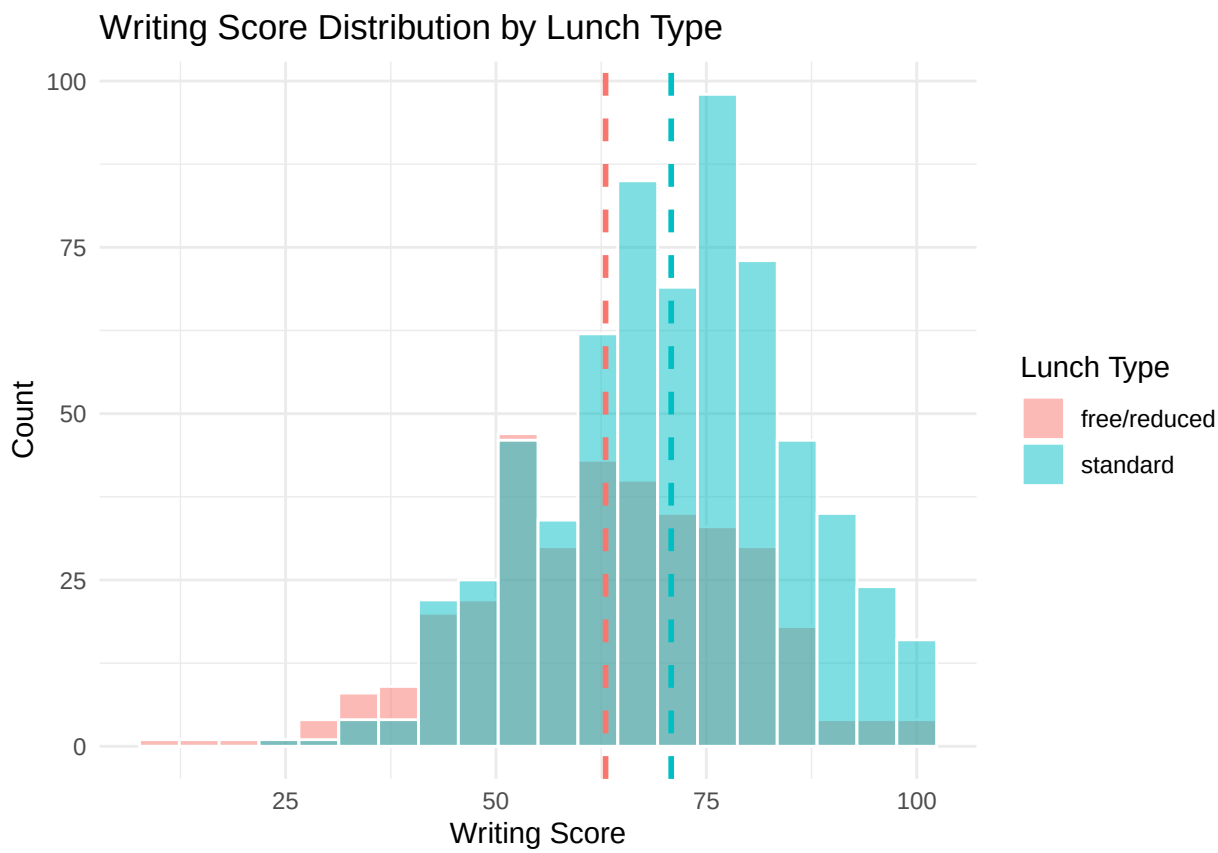
```
## [1] "Math score statistics by gender:"
```

```r
print(q7_stats)
```

```
## # A tibble: 2 x 6
##   gender  Mean Median    SD   Min   Max
##   <chr>  <dbl>  <dbl> <dbl> <int> <int>
## 1 female  63.6     65  15.5     0   100
## 2 male    68.7     69  14.4    27   100
```

##Question 8: Histogram - Writing Scores by Lunch Type

```r
# Create overlapping histograms
ggplot(students, aes(x = writing.score, fill = lunch)) +
  geom_histogram(bins = 20, position = "identity", alpha = 0.5, color = "white") +
  theme_minimal() +
  labs(title = "Writing Score Distribution by Lunch Type",
       x = "Writing Score",
       y = "Count",
       fill = "Lunch Type") +
  geom_vline(data = students %>%
               group_by(lunch) %>%
               summarise(mean_write = mean(writing.score)),
             aes(xintercept = mean_write, color = lunch),
             linetype = "dashed", size = 1,
             show.legend = FALSE)
```




```
# Summary statistics
q8_stats <- students %>%
  group_by(lunch) %>%
  summarise(
    Mean = mean(writing.score),
    Median = median(writing.score),
    SD = sd(writing.score)
  )

print("Writing score statistics by lunch type:")
```

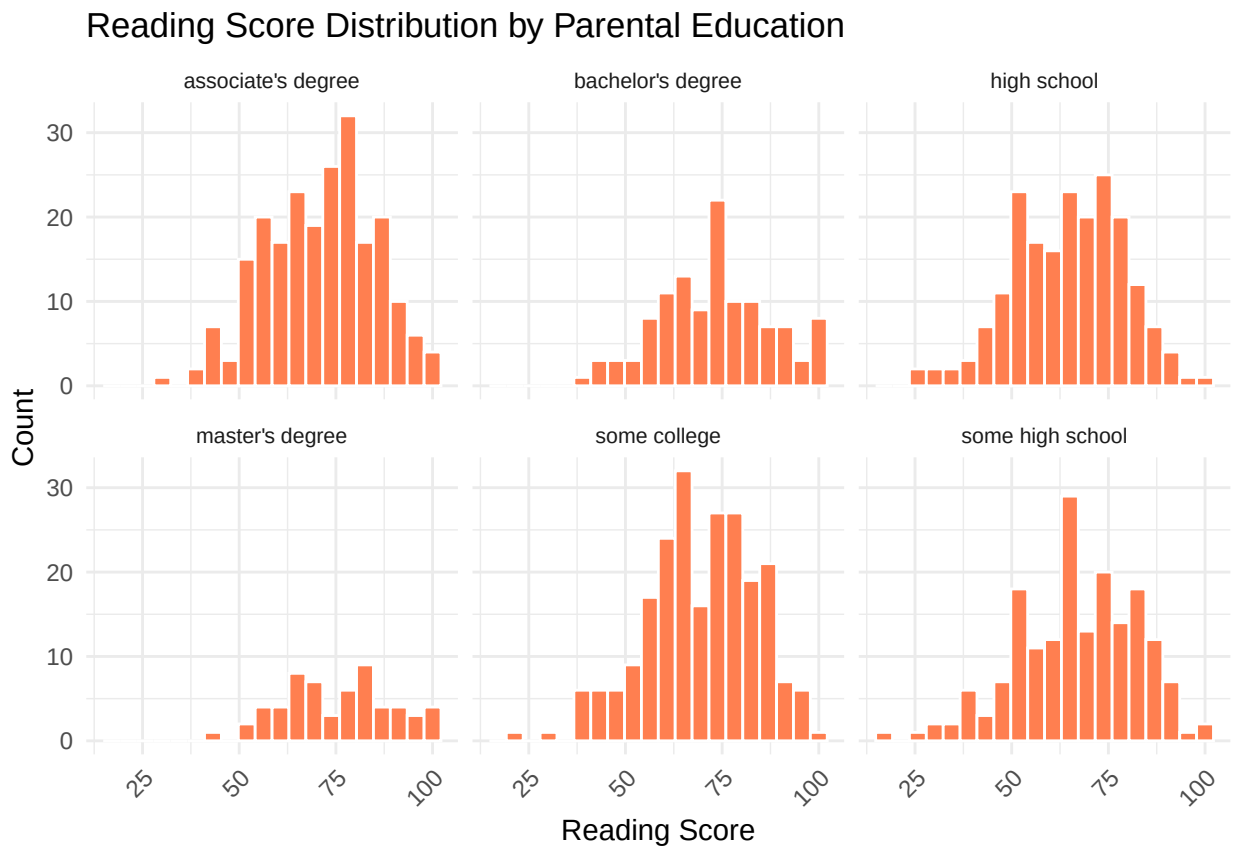
```
## [1] "Writing score statistics by lunch type:"
```

```
print(q8_stats)
```

```
## # A tibble: 2 x 4
##   lunch         Mean Median    SD
##   <chr>        <dbl>  <int> <dbl>
## 1 free/reduced  63.0     64  15.4
## 2 standard      70.8     72  14.3
```

##Question 9: Faceted Histogram - Reading Scores by Parental Education

```r
# Create faceted histogram
ggplot(students, aes(x = reading.score)) +
  geom_histogram(bins = 20, fill = "coral", color = "white") +
  facet_wrap(~parental.level.of.education, ncol = 3) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 8)) +
  labs(title = "Reading Score Distribution by Parental Education",
       x = "Reading Score",
       y = "Count")
```




```r
# Calculate mean and median to assess skewness
q9_skew <- students %>%
  group_by(parental.level.of.education) %>%
  summarise(
    Mean = mean(reading.score),
    Median = median(reading.score),
    Count = n()
  ) %>%
  mutate(
    Skewness = ifelse(Mean > Median, "Right-skewed",
                      ifelse(Mean < Median, "Left-skewed", "Symmetric"))
  ) %>%
  arrange(desc(Mean - Median))
```

```r
print("Skewness assessment:")
```

```
## [1] "Skewness assessment:"
```

```r
print(q9_skew)
```

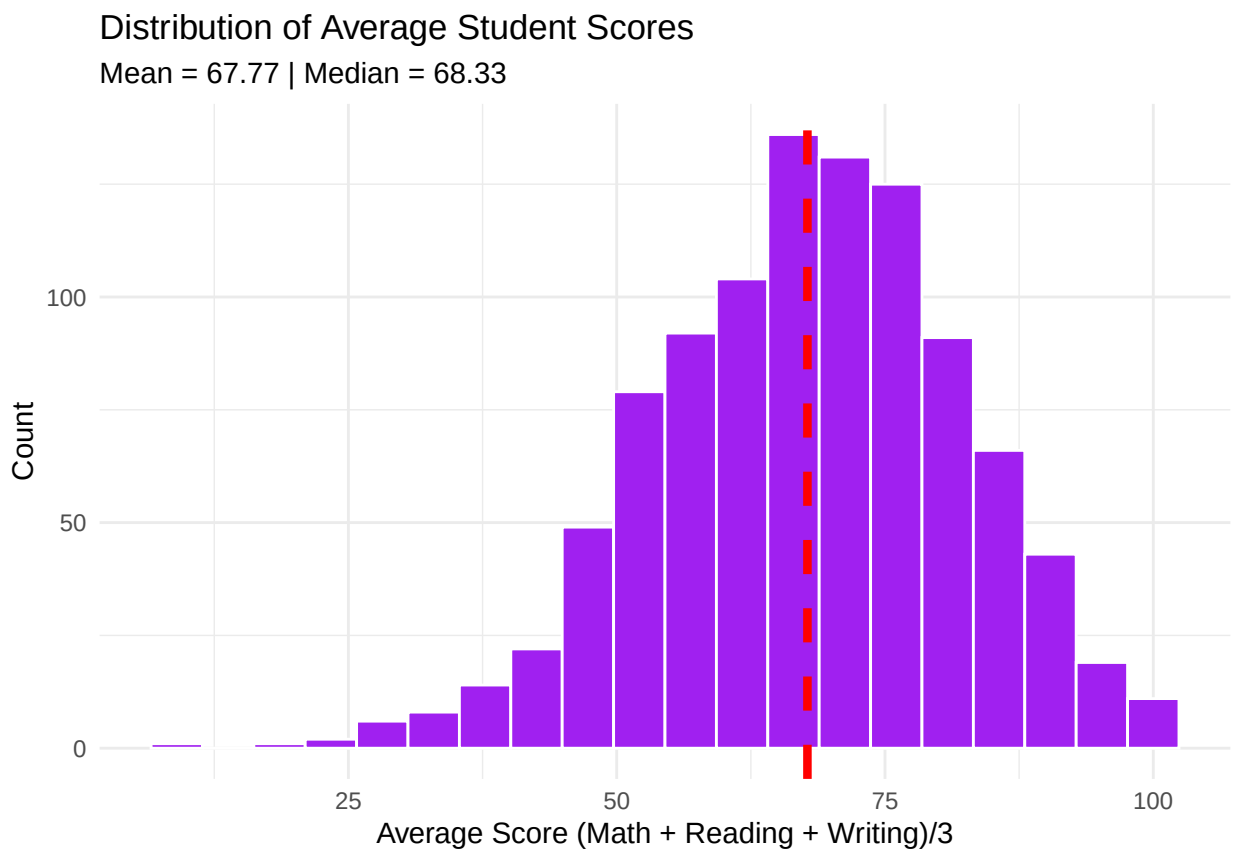
```
## # A tibble: 6 x 5
##   parental.level.of.education  Mean Median Count Skewness
##   <chr>                      <dbl>  <dbl> <int> <chr>
## 1 bachelor's degree           73     73     118 Symmetric
## 2 some high school            66.9   67     179 Left-skewed
## 3 master's degree             75.4   76      59 Left-skewed
## 4 some college                69.5   70.5   226 Left-skewed
## 5 high school                 64.7   66     196 Left-skewed
## 6 associate's degree          70.9   72.5   222 Left-skewed
```

##Question 10: Histogram - Average Student Score

```r
# Create average score variable
students <- students %>%
  mutate(average_score = (math.score + reading.score + writing.score) / 3)

# Calculate mean
mean_avg <- mean(students$average_score)

# Create histogram
ggplot(students, aes(x = average_score)) +
  geom_histogram(bins = 20, fill = "purple", color = "white") +
  geom_vline(xintercept = mean_avg,
             color = "red", size = 1.5, linetype = "dashed") +
  theme_minimal() +
  labs(title = "Distribution of Average Student Scores",
       subtitle = paste("Mean =", round(mean_avg, 2),
                        "| Median =", round(median(students$average_score), 2)),
       x = "Average Score (Math + Reading + Writing)/3",
       y = "Count")
```



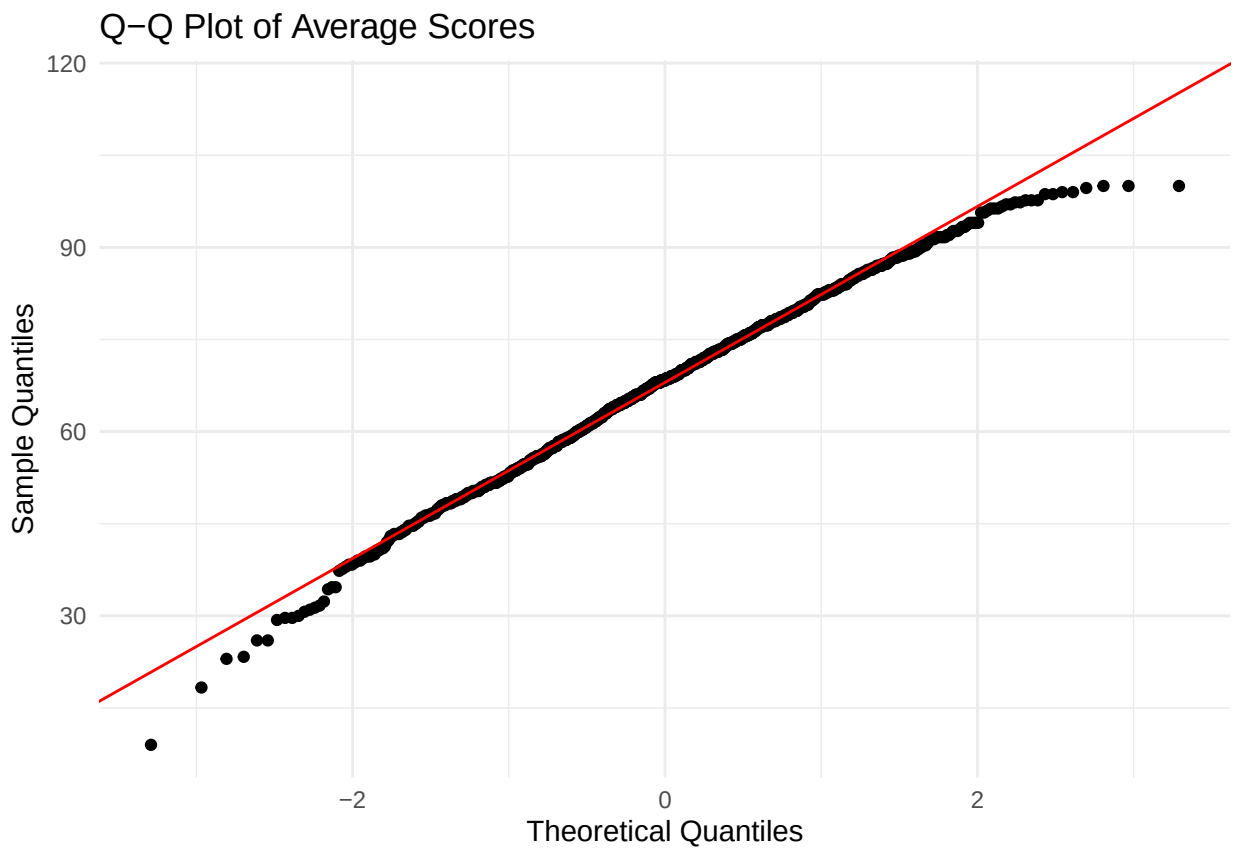

```r
# Summary statistics
q10_stats <- students %>%
  summarise(
    Mean = mean(average_score),
    Median = median(average_score),
    SD = sd(average_score),
    Min = min(average_score),
    Max = max(average_score)
  )

print("Average score statistics:")
```

```
## [1] "Average score statistics:"
```

```r
print(q10_stats)
```

```
##       Mean   Median       SD Min Max
## 1 67.77067 68.33333 14.25733   9 100
```

```r
# QQ plot for normality check
ggplot(students, aes(sample = average_score)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  theme_minimal() +
```

```r
labs(title = "Q-Q Plot of Average Scores",
     x = "Theoretical Quantiles",
     y = "Sample Quantiles")
```